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Subject: ADBMS

Experiment-05

Aim:

(A) A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion.

For each registered user, calculate the confirmation rate, which is defined as:

$$\text{Confirmation Rate} = \frac{\text{Number of Successful Confirmations}}{\text{Total Confirmation Requests}}$$

If a user has never received any confirmation request, their confirmation rate should be considered **0**.

Display the user ID along with the confirmation rate rounded to **2 decimal places**. The result can be returned in any order.

(B) Problem Statement

A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants.

For each day between **2013-10-01** and **2013-10-03**, determine the proportion of canceled trips among all valid trips. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the date and the calculated cancellation rate rounded to two decimal places.

We need to calculate:	But only consider trips where:	Date between
Cancellation Rate	Client is NOT banned	2013-10-01
=	AND	And
	Driver is NOT banned	2013-10-03
Cancelled Trips		

Total Valid Trips		

Tools / Platform Used:

- Windows
- PostgreSQL
- PgAdmin

Code and Output:

A.

```
CREATE TABLE Signups
(
    user_id INT PRIMARY KEY,
    time_stamp TIMESTAMP
);

CREATE TABLE Confirmations
(
    user_id INT,
    time_stamp TIMESTAMP,
    action VARCHAR(20),
    PRIMARY KEY(user_id, time_stamp),
    FOREIGN KEY(user_id)
    REFERENCES Signups(user_id)
);

INSERT INTO Signups
VALUES
(3, '2020-03-21 10:16:13'),
(7, '2020-01-04 13:57:59'),
(2, '2020-07-29 23:09:44'),
(6, '2020-12-09 10:39:37');

INSERT INTO Confirmations
VALUES
(3, '2021-01-06 03:30:46', 'timeout'),
(3, '2021-07-14 14:00:00', 'timeout'),
(7, '2021-06-12 11:57:29', 'confirmed'),
(7, '2021-06-13 12:58:28', 'confirmed'),
(7, '2021-06-14 13:59:27', 'confirmed'),
(2, '2021-01-22 00:00:00', 'confirmed'),
(2, '2021-02-28 23:59:59', 'timeout');
```

```

34
35 select S.user_id , round(avg(
36     CASE
37         WHEN C.action='confirmed' THEN 1
38         ELSE 0
39     END
40 ),2) as confirmation_rate
41 FROM Signups AS S
42 LEFT JOIN Confirmations AS C
43 ON S.user_id = C.user_id
44 GROUP BY S.user_id;

```

Data Output Messages Notifications

	user_id [PK] integer	confirmation_rate numeric
1	2	0.50
2	6	0.00
3	3	0.00
4	7	1.00

B.

```

INSERT INTO users
VALUES
(1,'No','client'),
(2,'Yes','client'),
(3,'No','client'),
(4,'No','client'),
(10,'No','driver'),
(11,'No','driver'),
(12,'No','driver'),
(13,'No','driver');

INSERT INTO Trips
VALUES
(1,1,10,1,'completed','2013-10-01'),
(2,2,11,1,'cancelled_by_driver','2013-10-01'),
(3,3,12,6,'completed','2013-10-01'),
(4,4,13,6,'cancelled_by_client','2013-10-01'),
(5,1,10,1,'completed','2013-10-02'),
(6,2,11,6,'completed','2013-10-02'),
(7,3,12,6,'completed','2013-10-02'),
(8,2,12,12,'completed','2013-10-03'),
(9,3,10,12,'completed','2013-10-03'),
(10,4,13,12,'cancelled_by_driver','2013-10-03');

```

```

42 SELECT
43     T.REQUEST_AT AS DAY,
44     ROUND(AVG(
45         CASE
46             WHEN T.STATUS IN
47                 ('cancelled_by_driver','cancelled_by_client')
48             THEN 1
49             ELSE 0
50             END
51         ),2)
52 FROM Trips t
53 JOIN Users c
54 ON t.client_id = c.users_id
55 JOIN Users d
56 ON t.driver_id = d.users_id
57 WHERE c.banned = 'No'
58     AND d.banned = 'No'
59     AND t.request_at BETWEEN
60         '2013-10-01'
61         AND
62         '2013-10-03'
63 GROUP BY T.REQUEST_AT;
64

```

Data Output Messages Notifications








SQL

Show

	day date	round numeric
1	2013-10-...	0.33
2	2013-10-...	0.00
3	2013-10-...	0.50